

Knowledge Distillation with Ternary Weight Quantization-Aware Training: An Experimental Study of Training Strategies

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ATDL Assignment 1

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Abstract—This work investigates the combination of Knowledge Distillation (KD) and Ternary Weight Networks (TWN) for training a compact ResNet-18 model. A ResNet-34 teacher provides the distillation signal, while the student uses ternary weights of the form $\{-\alpha, 0, +\alpha\}$. Four training strategies are compared to study the effect of TWN quantization-aware training (QAT), Straight-Through Estimation (STE), persistent full-precision latent weights, and two-stage versus joint training. The models are evaluated on CIFAR-10 using classification accuracy, sparsity, and storage requirements, with an additional evaluation on CIFAR-10.1. The best TWN configuration achieves 96.15% CIFAR-10 test accuracy, compared with 96.21% for the uncompressed FP32 ResNet-18 student, while reducing the reported model storage from 42.63 MB to 2.72 MB. The LAtE extension achieves 95.91% test accuracy with 66.85% overall sparsity, providing a different accuracy–sparsity trade-off.

Index Terms—knowledge distillation, ternary weight networks, quantization-aware training, model compression, straight-through estimator, ResNet

I. INTRODUCTION

Deep neural networks achieve strong predictive performance but can require substantial memory and computation. This creates challenges for deployment on resource-constrained hardware. **Model compression** aims to reduce these requirements while maintaining useful model performance.

Two relevant compression approaches are **Knowledge Distillation (KD)** [2] and **weight quantization**. KD transfers information from a larger teacher network to a smaller student network through the teacher’s output distribution. Weight quantization reduces the precision used to represent model parameters. In a **Ternary Weight Network (TWN)** [3], [4], each weight is represented as

$$\hat{W}_i = \alpha B_i, \quad B_i \in \{-1, 0, +1\}, \quad (1)$$

giving the ternary set $\{-\alpha, 0, +\alpha\}$.

The main difficulty is that ternary weights are discrete, while neural network optimization is performed using gradients. **Quantization-Aware Training (QAT)** addresses this by using quantized weights during the forward computation while maintaining trainable full-precision parameters. The **Straight-Through Estimator (STE)** [5], [6] provides an approximate gradient through the quantization operation.

This work studies the combination of KD and TWN-QAT using a ResNet-34 teacher and ResNet-18 student [1] on CIFAR-10. Different training strategies are compared to examine the effect of the KD signal, STE, latent full-precision parameters, and the timing of quantization on the resulting ternary model. LAtE is also evaluated as an additional extension of the TWN study rather than as the primary focus of the assignment. The complete codebase, experimental logs, and checkpoints are available at https://github.com/vaidhav-adit/Hybrid_KD_and_TWN_QAT. GPT Chat Link: <https://chatgpt.com/share/6aaad98e-65e0-83e8-9c5e-e9bb9a4535a5>

II. KD-TWN PROBLEM FORMULATION

The training setup combines a **knowledge-distillation objective** with a **ternary quantization mechanism**. The teacher provides the distillation signal, while the student performs its forward computation using ternary weights.

A. Teacher and Student

Let the teacher network be denoted by f_T and the student network by f_S . For an input image x , their output logits are

$$v = f_T(x), \quad z = f_S(x), \quad (2)$$

where $v, z \in \mathbb{R}^C$ and C is the number of classes.

The teacher is trained separately and kept fixed during distillation. The student is optimized using both the ground-truth label and the teacher’s output distribution.

B. Knowledge Distillation

The logits are converted into softened probability distributions using a temperature-scaled softmax. For the teacher,

$$p_i^{(T)} = \frac{\exp(v_i/T)}{\sum_{j=1}^C \exp(v_j/T)}, \quad (3)$$

and for the student,

$$q_i^{(T)} = \frac{\exp(z_i/T)}{\sum_{j=1}^C \exp(z_j/T)}, \quad (4)$$

where T is the **distillation temperature** and i indexes the classes.

The distillation loss is the cross-entropy between the softened teacher and student distributions:

$$L_{\text{KD}} = - \sum_{i=1}^C p_i^{(T)} \log q_i^{(T)}. \quad (5)$$

The student is also trained on the ground-truth class label using standard unscaled student probabilities $q_i^{(1)} = \exp(z_i) / \sum_j \exp(z_j)$:

$$L_{\text{CE}} = - \log q_y^{(1)}, \quad (6)$$

where y is the correct class label.

The two losses are combined as

$$L = \lambda T^2 L_{\text{KD}} + (1 - \lambda) L_{\text{CE}}, \quad (7)$$

with default settings

$$T = 4, \quad \lambda = 0.5. \quad (8)$$

The T^2 factor follows the standard KD formulation and compensates for the reduction in gradient magnitude caused by temperature scaling.

C. Ternary Weight Representation

For a full-precision weight tensor W , TWN constructs a ternary approximation

$$\hat{W} = \alpha B, \quad (9)$$

where $B_i \in \{-1, 0, +1\}$. The implementation determines the ternary state using a threshold Δ :

$$B_i = \begin{cases} +1, & W_i > \Delta, \\ 0, & |W_i| \leq \Delta, \\ -1, & W_i < -\Delta. \end{cases} \quad (10)$$

The threshold used in the experiments is

$$\Delta = 0.75 \mathbb{E}[|W|]. \quad (11)$$

The scaling factor is computed from the mean absolute magnitude of the weights assigned to the non-zero states:

$$\alpha = \frac{\sum_i |W_i| \mathbb{I}(|W_i| > \Delta)}{\sum_i \mathbb{I}(|W_i| > \Delta)}. \quad (12)$$

D. Combining KD with TWN-QAT

The student uses the quantized weights during the forward computation. Let W denote the persistent full-precision latent weights. The quantized weights are

$$W_q = Q_{\text{TWN}}(W), \quad (13)$$

where $Q_{\text{TWN}}(\cdot)$ is the ternary quantization function.

The student therefore produces

$$z = f_S(x; W_q). \quad (14)$$

The resulting logits are used to calculate both the KD and Cross-Entropy losses:

$$L = 0.5 T^2 L_{\text{KD}} + 0.5 L_{\text{CE}}. \quad (15)$$

The forward computation uses W_q , while the optimizer updates the persistent full-precision weights W . Since the ternary quantization operation is non-differentiable, the **Straight-Through Estimator** provides an approximate gradient:

$$\frac{\partial L}{\partial W} \approx \frac{\partial L}{\partial W_q}. \quad (16)$$

The latent weights are then updated using

$$W_{t+1} = W_t - \eta \frac{\partial L}{\partial W_t}, \quad (17)$$

where η is the learning rate.

The resulting training flow can be represented compactly as

$$W \xrightarrow{Q_{\text{TWN}}} W_q \xrightarrow{f_S} z \xrightarrow{L} \text{loss} \xrightarrow{\text{STE}} W. \quad (18)$$

This persistent full-precision parameter is an important distinction between TWN-QAT and the direct hard-projection strategy evaluated later.

III. LOSS-AWARE TERNARIZATION

The TWN formulation described above determines the ternary approximation by minimizing the reconstruction error between the full-precision weights and their ternary representation. However, this objective treats all weights equally. **Loss-Aware Ternarization (LATE)**, introduced by Hou and Kwok [7], extends this formulation by incorporating an estimate of the loss sensitivity of individual weights.

The important difference is not the ternary representation itself. Instead, LATE changes the **objective used to find the ternary approximation**.

A. Loss-Aware Objective

Let W denote the full-precision weight vector and let $\hat{W} = \alpha B$ denote its ternary approximation. LATE introduces a weight-dependent sensitivity term d_i and minimizes the weighted reconstruction error:

$$\min_{\alpha, B} \frac{1}{2} \sum_{i=1}^n d_i (\alpha B_i - W_i)^2. \quad (19)$$

Here, d_i represents the estimated sensitivity associated with weight W_i . A larger value of d_i gives greater importance to the reconstruction error of that weight, while a smaller value gives it less influence.

Therefore, instead of considering only how closely the original weights can be reconstructed, LATE also considers the relative importance of different reconstruction errors.

B. Optimal LATe Quantization

For a fixed α , minimizing the weighted objective with respect to each ternary state gives

$$B_i = \begin{cases} +1, & W_i > \frac{\alpha}{2}, \\ 0, & |W_i| \leq \frac{\alpha}{2}, \\ -1, & W_i < -\frac{\alpha}{2}. \end{cases} \quad (20)$$

Thus, the corresponding optimal threshold is

$$\Delta^* = \frac{\alpha}{2}. \quad (21)$$

For a fixed ternary assignment B , the optimal scaling factor is

$$\alpha^* = \frac{\sum_i d_i |W_i| \mathbb{I}(B_i \neq 0)}{\sum_i d_i \mathbb{I}(B_i \neq 0)}. \quad (22)$$

Unlike the standard TWN calculation, the contribution of each non-zero weight is weighted by its corresponding d_i .

C. Exact LATe Solver

The implementation used in this work follows the exact symmetric LAT formulation of Hou and Kwok [7]. The absolute weight magnitudes are sorted in descending order:

$$|W_{(1)}| \geq |W_{(2)}| \geq \dots \geq |W_{(n)}|. \quad (23)$$

For a candidate partition containing the first k sorted weights, the optimal scaling factor is

$$\alpha_k = \frac{\sum_{j=1}^k d_{(j)} |W_{(j)}|}{\sum_{j=1}^k d_{(j)}}. \quad (24)$$

The weighted objective is evaluated for each candidate partition. The implementation uses cumulative prefix sums to evaluate these candidates efficiently and selects the partition with the minimum weighted reconstruction error.

If k^* is the selected partition, the resulting threshold is

$$\Delta^* = \frac{\alpha_{k^*}}{2}. \quad (25)$$

Thus, LATe performs an **exact search over possible ternary partitions** rather than using the fixed heuristic threshold used by TWN in this work.

D. Difference from TWN

The difference between TWN and LATe can be expressed directly through their optimization objectives. TWN uses an unweighted reconstruction error,

$$L_{\text{TWN}} = \sum_i (\alpha B_i - W_i)^2, \quad (26)$$

whereas LATe uses

$$L_{\text{LATe}} = \sum_i d_i (\alpha B_i - W_i)^2. \quad (27)$$

The ternary representation remains the same in both methods. The difference is the presence of the weight-dependent factor d_i .

If the curvature is identical for every weight,

$$d_i = \lambda, \quad \forall i, \quad (28)$$

then λ is a common factor and does not affect the minimizer. The LATe objective consequently reduces to the TWN objective:

$$D = \lambda I \implies \text{LATe} \equiv \text{TWN}. \quad (29)$$

The practical distinction therefore comes from allowing different weights to have different values of d_i .

E. Sensitivity Estimate in This Work

In our implementation, the sensitivity term is estimated from squared gradients:

$$d_i \approx \mathbb{E}[g_i^2], \quad (30)$$

where

$$g_i = \frac{\partial L}{\partial W_i}. \quad (31)$$

This provides a practical estimate of weight sensitivity for the loss-aware objective. LATe is included as an **additional study** to examine whether incorporating loss sensitivity changes the accuracy–sparsity trade-off. The primary focus of this assignment remains Knowledge Distillation and TWN-based QAT.

IV. EXPERIMENTAL DESIGN

The experiments are designed around a common ResNet-18 student and a frozen ResNet-34 teacher. The main purpose is to isolate the effect of different **training strategies** rather than changing the model architecture between experiments.

A full-precision ResNet-18 trained only with supervised Cross-Entropy is first used as a reference model. The four main ternary baselines then introduce TWN, Knowledge Distillation, hard projection, and QAT in a controlled sequence. The resulting framework is followed by the primary baseline comparison and two controlled ablation studies: a **distillation temperature ablation** and a **first/last-layer precision ablation**. Finally, an additional LATe study examines a loss-aware alternative to the standard TWN quantization objective.

A. Common Experimental Setup

All primary experiments use CIFAR-10 with the same ResNet-18 student architecture and training data split. The ResNet-34 teacher is trained separately using full-precision weights and is kept frozen during Knowledge Distillation.

Unless otherwise stated, the experiments use a distillation temperature of $T = 4$, KD weight $\lambda = 0.5$, and the TWN threshold

$$\Delta = 0.75 \mathbb{E}[|W|]. \quad (32)$$

The total training budget is 200 epochs. SGD with momentum (0.9), weight decay (5×10^{-4}), and a cosine learning-rate schedule with a 5-epoch linear warmup are used across the experiments. The supervised Cross-Entropy loss uses a label-smoothing factor of 0.1 throughout all training runs. The random seed is fixed to 42 for reproducibility.

The standard TWN configuration keeps the first convolutional layer and the final fully connected layer in FP32, while the intermediate convolutional layers use ternary weights. Batch-normalization parameters remain in FP32. The first/last-layer ablation later quantizes these two boundary layers as well.

B. Four Baselines

The four baselines are designed to progressively examine the role of **ternary quantization**, **Knowledge Distillation**, **continuous latent weights**, and the **Straight-Through Estimator**.

1) *Baseline 1: TWN with Supervised Cross-Entropy*: Baseline 1 trains a TWN ResNet-18 from scratch using only the supervised Cross-Entropy loss on ground-truth labels. The student performs ternary forward computations, but no teacher information is provided.

$$L_{B1} = L_{CE}. \quad (33)$$

B1 demonstrates that the ternary ResNet-18 architecture can achieve 95.65% test accuracy under the evaluated CIFAR-10 training setup without Knowledge Distillation.

A separate FP32 ResNet-18 trained with supervised Cross-Entropy is used as the **Student Base Reference**. It is not counted as one of the four ternary baselines.

2) *Baseline 2: Two-Stage KD + TWN-QAT*: Baseline 2 uses a two-stage training strategy. In Stage 1, a full-precision ResNet-18 student is trained using Knowledge Distillation from the frozen teacher. After this warmup, the learned full-precision parameters are used to initialize the TWN-QAT student for Stage 2.

The two stages can be represented as

$$\text{FP32 KD Warmup} \rightarrow \text{TWN-QAT Fine-Tuning}. \quad (34)$$

The second stage uses the combined KD and Cross-Entropy objective defined in Section II. This baseline tests whether reaching a useful full-precision solution before ternarization provides an advantage.

3) *Baseline 3: Direct Hard-Projected Ternary Training*: Baseline 3 directly trains a ternary ResNet-18 from scratch while using the KD objective. After each optimizer update, the parameters are projected back to the ternary representation:

$$W_{t+1} = Q_{\text{TWN}}(W_t - \eta \nabla_W L). \quad (35)$$

Unlike QAT, this strategy does not maintain a separate persistent full-precision latent parameter through the ternary projection and does not use the STE mechanism.

This baseline acts as a **negative control**. Its purpose is to compare direct discrete parameter updates against the latent-weight optimization used by QAT. Any explanation for differences in performance is treated as a hypothesis and is evaluated through the observed results.

4) *Baseline 4: Joint KD + TWN-QAT*: Baseline 4 performs Knowledge Distillation and TWN-QAT jointly from the first training epoch. The model maintains full-precision latent weights W , while the forward pass uses the dynamically generated ternary weights W_q :

$$W_q = Q_{\text{TWN}}(W). \quad (36)$$

The combined loss is then computed from the student logits, and the STE provides the approximate gradient required to update the latent weights.

This is the main **end-to-end KD-QAT configuration** investigated in the assignment. Unlike Baseline 2, it does not require a separate full-precision KD warmup stage.

C. Purpose of the Baseline Comparison

The four baselines provide a controlled progression:

- **B1** establishes the ternary supervised baseline.
- **B2** studies progressive training through FP32 KD warmup followed by QAT.
- **B3** removes the persistent latent-weight and STE mechanism through direct hard projection.
- **B4** combines KD and QAT jointly from initialization.

The comparison therefore focuses on the training process rather than only the final compression ratio. In particular, B2, B3, and B4 allow the role of **training schedule** and **continuous latent optimization** to be examined separately.

D. Ablation Studies

Following the baseline comparison, two controlled ablation studies are performed: a distillation-temperature ablation and a first/last-layer precision ablation. The LATe experiment is treated separately as an additional study.

1) *Distillation Temperature Ablation*: The second study examines the effect of the KD temperature T . The temperature controls the softness of the teacher and student probability distributions:

$$p_i(T) = \frac{\exp(v_i/T)}{\sum_j \exp(v_j/T)}. \quad (37)$$

The experiments evaluate

$$T \in \{2, 4, 15\}.$$

The remaining training configuration is kept unchanged. This allows the effect of the **distillation temperature** on the ternary student to be examined without changing the quantization mechanism.

2) *First/Last-Layer Precision Ablation*: The third study examines whether the first convolutional stem and final linear classification layer should remain in full precision. The standard configuration maintains an FP32 stem and classifier with ternary intermediate residual layers:

$$\text{FP32 Stem} + \text{Ternary Residuals} + \text{FP32 Classifier.} \quad (38)$$

The fully quantized ablation constrains the entire network to ternary precision:

$$\text{Ternary Stem} + \text{Ternary Residuals} + \text{Ternary Classifier.} \quad (39)$$

This experiment evaluates the trade-off between complete network ternarization, model storage footprint, layer sparsity, and top-1 accuracy.

E. Additional LATE Study

The final experimental branch replaces the standard TWN quantization objective with the **Loss-Aware Ternarization (LATE)** formulation described in Section III. The network architecture and KD training strategy are otherwise kept comparable to the TWN experiments.

The purpose of this study is to examine whether incorporating weight-dependent loss sensitivity changes the resulting **accuracy-sparsity trade-off**. LATE is therefore treated as an additional quantization study rather than as one of the primary assignment baselines.

V. EXPERIMENTAL RESULTS AND DISCUSSION

This section presents the results from the primary baselines, controlled ablations, and the LATE extension. The evaluation covers **classification accuracy**, model sparsity, storage requirements, weight distributions, and performance on the unseen CIFAR-10.1 dataset.

A. Master Experimental Results

Table I summarizes the results across the reference models, four primary baselines, controlled ablations, and the LATE extension. All models were trained for 200 epochs on CIFAR-10 using a fixed random seed of 42 and evaluated on the official 10,000 image test split.

B. Analysis of Training Strategies

The primary baseline comparison highlights clear differences between the four training strategies.

- **Baseline 1**: TWN trained from scratch using supervised Cross-Entropy achieves 95.65% test accuracy, compared with 96.21% for the uncompressed FP32 Student Base. The model has 52.00% sparsity and a reported model storage of 2.72 MB.
- **Baseline 2**: The two-stage approach reaches 96.01% test accuracy. The FP32 KD warmup is followed by TWN-QAT, giving a solid performance recovery approaching the full-precision baseline.
- **Baseline 3**: Direct hard projection gives 82.01% test accuracy, substantially below the other ternary configurations. Its sparsity is also lower at 34.91%. This result

suggests that directly projecting the parameters to ternary values during optimization behaves differently from the latent-weight QAT approach.

- **Baseline 4**: Joint KD and TWN-QAT achieves the highest CIFAR-10 test accuracy among the primary baselines at 96.15%, effectively matching the 96.21% FP32 Student Base and approaching the 96.54% ResNet-34 Teacher. Unlike Baseline 2, it applies KD and QAT jointly from the beginning without a separate FP32 warmup stage.

These results indicate that the **training strategy** has a substantial effect on the behavior of the ternary student. In particular, the difference between B3 and the QAT-based configurations motivates further analysis of the role of latent full-precision parameters and STE. The experiments alone do not establish a single causal explanation for this difference.

C. Learning Dynamics and Training Curves

Figure 1 shows the training and validation curves for the uncompressed FP32 ResNet-18 Student Base. The model reaches a stable performance region during training under the cosine learning-rate schedule. These curves provide the reference training behavior for comparison with the quantized models.

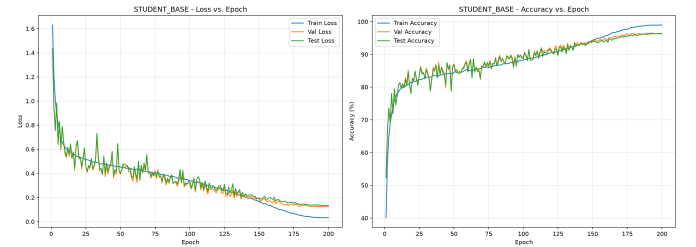


Fig. 1. Training and validation loss and accuracy curves for the FP32 ResNet-18 Student Base over 200 epochs.

D. Weight Distribution and Histogram Analysis

Figure 6 shows the weight distributions for the four default TWN baselines. The histograms provide two views of the weights: the latent FP32 distribution and the resulting ternary representation.

For Baselines 1, 2, and 4, the latent weights form continuous distributions around zero before quantization. The TWN threshold maps the weights within the dead zone to zero, resulting in 52.00% overall sparsity for these configurations.

The resulting non-zero states are represented by $-\alpha$ and $+\alpha$, giving the expected symmetric ternary structure. Baseline 3 differs because the model uses direct hard projection rather than a persistent latent FP32 parameter. Its resulting ternary distribution has lower overall sparsity at 34.91%.

These distributions describe the observed parameter structure. They do not by themselves establish why the corresponding training strategies produce different accuracies.

E. Controlled Ablation Studies

1) *Distillation Temperature*: The first ablation changes the KD temperature while keeping the other training settings fixed.

TABLE I

CONSOLIDATED RESULTS ACROSS THE REFERENCE MODELS, PRIMARY BASELINES, CONTROLLED ABLATIONS, AND LATE EXTENSION ON CIFAR-10.

Configuration	Precision	Strategy	T	First/Last	Val Acc. (%)	Test Acc. (%)	Sparsity (%)	Storage (MB)
<i>Reference Models (Unquantized FP32)</i>								
Teacher Reference (ResNet-34)	FP32	Supervised CE	N/A	FP32	97.88	96.54	0.00	85.27
Student Base Reference (ResNet-18)	FP32	Supervised CE	N/A	FP32	97.02	96.21	0.00	42.63
<i>Default TWN Baselines ($T = 4.0$, $\lambda = 0.5$, FP32 First/Last)</i>								
Baseline 1 (TWN Supervised)	Ternary	From Scratch (CE)	N/A	FP32	95.94	95.65	52.00	2.72
Baseline 2 (Two-Stage Warmup)	Ternary	FP32 KD $\rightarrow$ QAT	4.0	FP32	96.54	96.01	52.00	2.72
Baseline 3 (Hard Projection)	Ternary	In-Place KD Step	4.0	FP32	82.66	82.01	34.91	2.72
Baseline 4 (Joint KD + TWN-QAT)	Ternary	Joint Scratch + STE	4.0	FP32	96.48	96.15	52.00	2.72
<i>Ablation 1: Low Distillation Temperature ($T = 2.0$, $\lambda = 0.5$)</i>								
Ablation 1a – Baseline 2	Ternary	Two-Stage KD+QAT	2.0	FP32	96.24	95.90	52.00	2.72
Ablation 1a – Baseline 3	Ternary	Hard Projection	2.0	FP32	82.10	81.85	34.91	2.72
Ablation 1a – Baseline 4	Ternary	Joint KD+QAT (STE)	2.0	FP32	96.20	95.85	52.00	2.72
<i>Ablation 1b: High Distillation Temperature ($T = 15.0$, $\lambda = 0.5$)</i>								
Ablation 1b – Baseline 2	Ternary	Two-Stage KD+QAT	15.0	FP32	96.30	96.22	52.00	2.72
Ablation 1b – Baseline 3	Ternary	Hard Projection	15.0	FP32	82.32	82.19	34.91	2.72
Ablation 1b – Baseline 4	Ternary	Joint KD+QAT (STE)	15.0	FP32	96.38	96.02	52.00	2.72
<i>Ablation 2: Boundary Layer Precision (Full-Network Ternarization, $T = 4.0$)</i>								
Ablation 2 – Baseline 1	FP32	Supervised CE	N/A	Full FP32	96.00	95.60	0.00	42.63
Ablation 2 – Baseline 2	Full Ternary	Two-Stage KD+QAT	4.0	Ternary	96.38	95.71	60.15	2.70
Ablation 2 – Baseline 3	Full Ternary	Hard Projection	4.0	Ternary	78.20	78.21	33.10	2.70
Ablation 2 – Baseline 4	Full Ternary	Joint KD+QAT (STE)	4.0	Ternary	96.36	95.90	60.15	2.70
<i>Additional Study: Loss-Aware Ternarization (LATE, $T = 4.0$, $\lambda = 0.5$)</i>								
LATE – Baseline 1	FP32	Supervised CE	N/A	FP32	95.94	95.61	0.00	42.63
LATE – Baseline 2	LATE Ternary	Two-Stage KD+QAT	4.0	FP32	95.94	95.76	66.85	2.72
LATE – Baseline 3	LATE Ternary	Hard Projection	4.0	FP32	72.76	71.86	31.40	2.72
LATE – Baseline 4	LATE Ternary	Joint KD+QAT (STE)	4.0	FP32	96.54	95.91	66.85	2.72

The experiments evaluate temperatures across three distinct regimes: $T = 2$ (peaked soft targets), $T = 4$ (balanced default), and $T = 15$ (diffused soft targets).

At low temperature ($T = 2$, $T^2 = 4$), Baseline 4 reaches 95.85% test accuracy. At the default temperature ($T = 4$, $T^2 = 16$), Baseline 4 achieves 96.15%, which is the best-performing setting among the tested temperatures. At high temperature ($T = 15$, $T^2 = 225$), Baseline 4 achieves 96.02%, while Baseline 2 reaches 96.22%. Across the tested temperatures, Baseline 3 remains substantially below the QAT-based configurations. In this experiment, changing the distillation temperature does not eliminate the performance difference between the hard-projection and QAT-based strategies.

Figure 7 shows the training curves for Baseline 4 with $T = 15$.

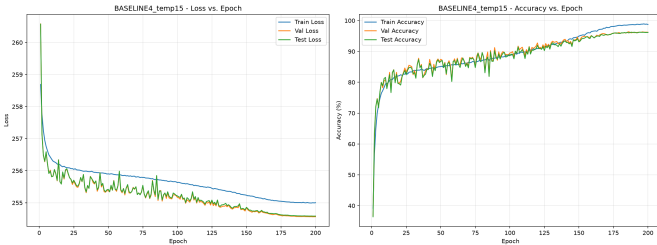


Fig. 7. Training and validation curves for Baseline 4 with distillation temperature $T = 15$.

2) *Boundary Layer Precision*: The second ablation quantizes the first convolutional layer and the final classification layer, which remain in FP32 in the standard configuration.

With full-network ternarization, overall sparsity increases from 52.00% to 60.15%, while the reported model storage decreases from 2.72 MB to 2.70 MB. The joint KD-QAT configuration achieves 95.90% test accuracy under this setting.

The accuracy difference between the standard and fully ternarized Baseline 4 configurations is 0.25 percentage points in this run. Thus, quantizing the boundary layers provides additional compression with a relatively small change in the measured classification accuracy.

Figure 8 shows the resulting weight distribution.

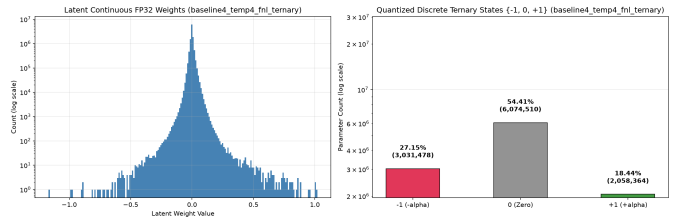


Fig. 8. Weight distribution for the fully quantized Baseline 4 model, including the first convolutional layer and final classifier.

F. Loss-Aware Ternarization Evaluation

The additional LATE study replaces the standard TWN reconstruction objective with the **loss-aware weighted objective** described in Section III. The resulting ternary model uses the same symmetric ternary representation but incorporates estimated weight sensitivity during the quantization step.

LATE Baseline 4 achieves 95.91% test accuracy with 66.85% overall sparsity. For comparison, the corresponding

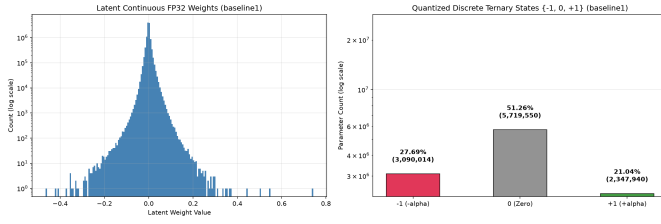


Fig. 2. *

(a) Baseline 1 (Supervised TWN, No KD)

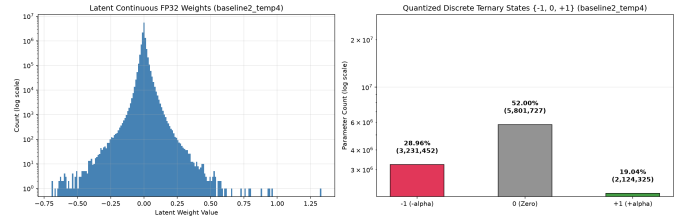


Fig. 3. *

(b) Baseline 2 (Two-Stage KD + TWN-QAT)

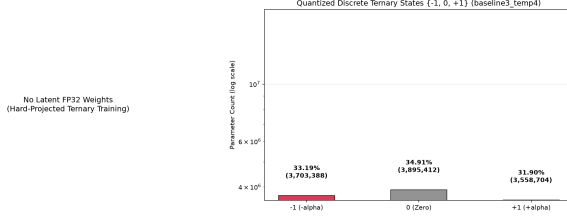


Fig. 4. *

(c) Baseline 3 (Hard Direct Projection)

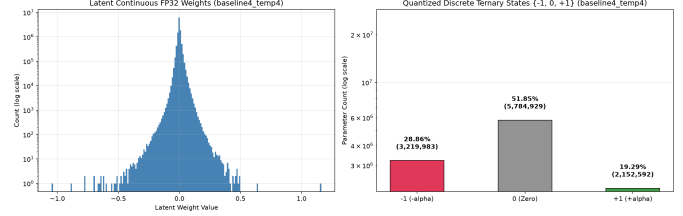


Fig. 5. *

(d) Baseline 4 (Joint KD + TWN-QAT)

Fig. 6. Weight distributions for the four default TWN baselines. The left view represents the latent FP32 weights, while the right view shows the ternary states $\{-1, 0, +1\}$.

TWN Baseline 4 achieves 96.15% accuracy with 52.00% sparsity.

This represents a higher sparsity level for LATe with a small difference in measured test accuracy. The result suggests a different accuracy–sparsity trade-off, although the single-run experiment does not establish that loss-aware quantization will produce the same behavior across other datasets or architectures.

TABLE II
LAYERWISE SPARSITY FOR TWN BASELINE 4 AND LATe BASELINE 4.

Layer	TWN (%)	LATe (%)
layer1.0.conv1	74.2	89.1
layer1.0.conv2	53.6	77.7
layer2.0.shortcut.0	54.8	87.7
layer2.1.conv2	52.2	66.5
layer3.1.conv2	53.9	71.0
layer4.1.conv2	55.3	74.9
Overall Mean	52.00	66.85

Table II shows that LATe produces higher sparsity across the listed layers. The largest differences occur in some early residual layers and shortcut projections, while the deeper layers retain a larger fraction of their weights.

Figure 11 presents the training curves and weight distribution for LATe Baseline 4. The histogram highlights the pronounced zero-weight peak produced by sensitivity-modulated thresholding, achieving 66.85% mean sparsity compared with 52.00% in standard TWN.

G. Evaluation on Unseen CIFAR-10.1

The trained models were additionally evaluated on the unseen CIFAR-10.1 benchmark. The results are shown in Table III.

TABLE III
EVALUATION ON THE UNSEEN CIFAR-10.1 BENCHMARK.

Model	CIFAR-10 (%)	CIFAR-10.1 (%)
ResNet-34 Teacher	96.54	90.80
ResNet-18 Student Base	96.21	89.75
Best TWN Student (B4)	96.15	90.95
Best LATe Student (B4)	95.91	89.90

The B4 TWN model achieves 90.95% accuracy on CIFAR-10.1, compared with 89.75% for the FP32 Student Base. The LATe B4 model achieves 89.90%. Thus, the TWN B4 configuration retains competitive accuracy on the unseen dataset despite its reduced weight precision.

Figure 12 shows the confusion matrices for the evaluated models.

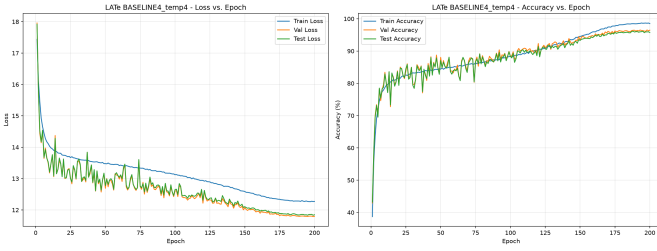


Fig. 9. *

(a) LATE Baseline 4 Learning Curves

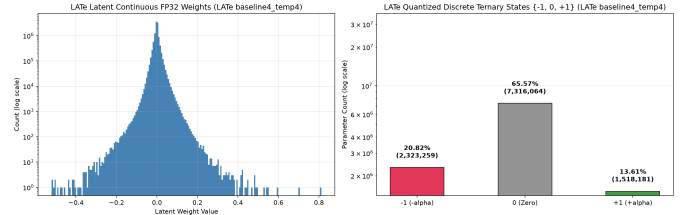


Fig. 10. *

(b) LATE Baseline 4 Weight Distribution

Fig. 11. Training curves and ternary weight distributions for Loss-Aware Ternarization (LATE) Baseline 4.

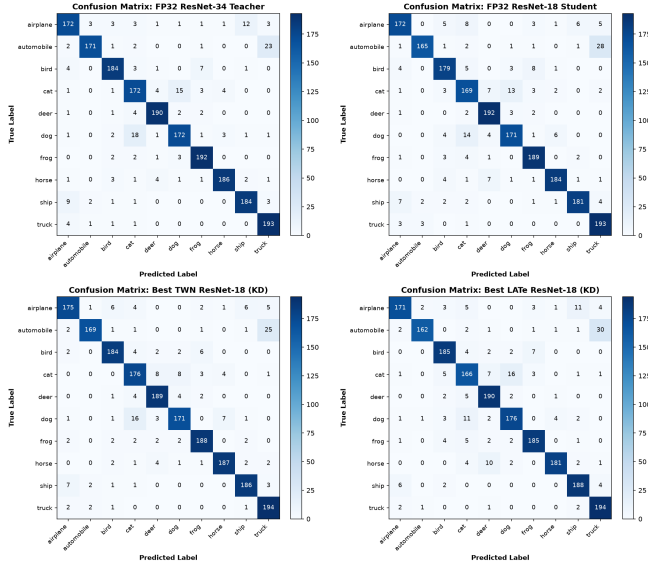


Fig. 12. Confusion matrices for the Teacher, Student Base, TWN Baseline 4, and LATE Baseline 4 on CIFAR-10.1.

H. Hardware Execution and Compression Potential

The compression and sparsity results for the main configurations are summarized in Table IV.

TABLE IV
STORAGE AND SPARSITY ACROSS DISTINCT MODEL CONFIGURATIONS.

Model Configuration	Sparsity (%)	Storage (MB)	Reduction
FP32 Student Base Reference	0.00	42.63	1.0×
Baseline 1 (TWN Supervised)	52.00	2.72	15.7×
Baseline 2 (Two-Stage KD+TWN)	52.00	2.72	15.7×
Baseline 3 (Hard Projection)	34.91	2.72	15.7×
Baseline 4 (Joint KD + TWN-QAT)	52.00	2.72	15.7×
Baseline 4 (Full Ternary FNL)	60.15	2.70	15.8×
Baseline 4 (LATE Loss-Aware)	66.85	2.72	15.7×

The standard TWN configurations reduce reported model storage from 42.63 MB for the FP32 Student Base to 2.72 MB, corresponding to approximately a 15.7× reduction. LATE increases overall sparsity from 52.00% to 66.85% under the same reported storage footprint.

I. Experimental Scope and Code Availability

All reported results correspond to a single deterministic run with random seed 42. Therefore, the observed differences should be interpreted as empirical results for this experimental setting rather than evidence of statistically significant performance differences across random initializations. The complete implementation, trained checkpoints, evaluation logs, and visual generation scripts are available for verification at https://github.com/vaidhav-adit/Hybrid_KD_and_TWN_QAT.

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